

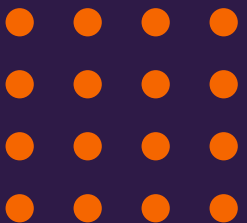


Outlier Detection & Analysis

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Outlier Analysis - Outline

Introduction / Motivation / Definition

Statistical-based Detection

- Distribution-based, depth-based

Deviation-based Detection

- Sequential exception, OLAP data cube

Distance-based Detection

- Index-based, nested-loop, cell-based, local-outliers

Questions



Introduction

Traditional Data Mining Categories

- Majority of Objects
 - Dependency detection
 - Class identification
 - Class description
- Exceptions
 - Exception/outlier detection

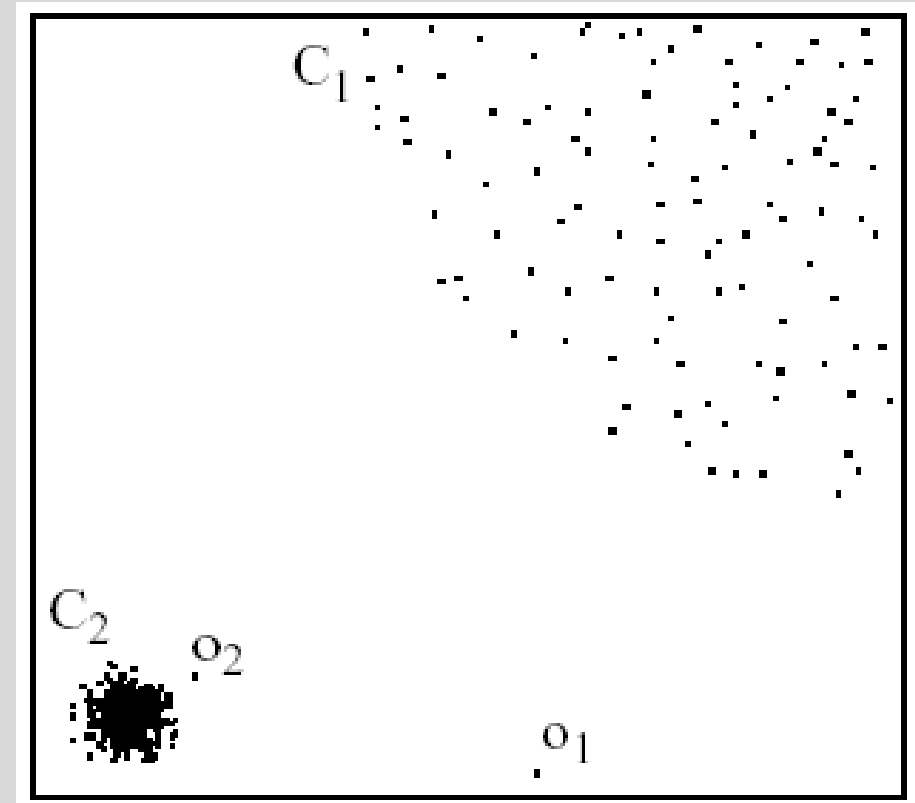
Motivation for Outlier Analysis

- Fraud Detection
(Credit card, telecommunications, criminal activity in e-Commerce)
- Customized Marketing (high/low income buying habits)
- Medical Treatments (unusual responses to various drugs)
- Analysis of performance statistics (professional athletes)
- Weather Prediction
- Financial Applications (loan approval, stock tracking)

“One persons noise could be another person’s signal.”

What is an outlier?

- Observations inconsistent with rest of the dataset – Global Outlier
- Special outliers – Local Outlier
 - Observations inconsistent with their neighborhoods
 - A local instability or discontinuity





Causes of Outliers

- Poor data quality / contamination
- Low quality measurements
- Malfunctioning equipment
- Manual error
- Correct but exceptional data
- Many more!



Outlier Detection Approaches

Objective:

- Define what data can be considered as inconsistent in a given data set
 - Statistical-Based Outlier Detection
 - Deviation-Based Outlier Detection
 - Distance-Based Outlier Detection
- Find an efficient method to mine the outliers



Why A Special Technique to Identify Outliers?

Why not just modify clustering or other algorithms to detect outliers?

- Performance considerations
- Subjective to the clustering algorithm and clustering parameters
- Only certain attributes may have outlier properties, no need to disqualify the entire tuple
- Contamination may occur by “column”, not by row